

AI Programming Foundations Project

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Overview

This project implemented a reproducible data analysis workflow using Python to explore the New York City Airbnb Listings dataset. The workflow included loading the dataset, cleaning and transforming the data, performing exploratory analysis, and creating visuals to identify patterns in Airbnb pricing and listing distribution. This project shows that using structured data workflows helps make data science practices more transparent and reproducible.

Dataset Description

The dataset used in this project is the NYC Airbnb Listings dataset obtained from Kaggle. The dataset contains information about Airbnb listings across New York City, including the price, room type, geographic location, number of reviews, and availability information. The dataset includes approximately 48,895 rows and 16 columns representing thousands of Airbnb listings across Manhattan, Brooklyn, Queens, the Bronx, and Staten Island. Key variables examined in this project included price, neighbourhood_group, number_of_reviews, and reviews_per_month, which provide insight into pricing patterns and listing popularity.

Workflow Description

The analysis used a structured data science workflow with multiple stages. First, the dataset was ingested using the Pandas library and loaded into a DataFrame for analysis. Next, data cleaning functions were implemented to address missing values and remove unrealistic price outliers. After cleaning the dataset, exploratory data analysis was performed using summary statistics and correlation analysis to better understand the relationships between variables. Visuals were then created using Matplotlib and Seaborn to explore price distributions, listing concentrations across boroughs, and relationships between price and review activity. This

structured workflow supports reproducibility by integrating code, data analysis, and narrative explanation within a single computational notebook. Reproducible workflows are an important component of modern data science because they allow analyses to be replicated and verified by others (Danchev, 2022).

Key Decisions and Assumptions

Several decisions were made during the data cleaning and analysis process. Missing values in the `reviews_per_month` column were replaced with zeros because listings without reviews likely indicate no recorded review activity rather than missing information. Rows with missing listing names or host names were removed because they represent incomplete records.

In addition, listings with prices below \$10 or above \$1000 were removed to reduce the influence of extreme outliers that could distort visualizations and summary statistics. These decisions were made to ensure the dataset more accurately represents typical Airbnb pricing patterns. During exploratory analysis, visuals were designed to highlight price distribution, geographic listing patterns, and relationships between price and user engagement.

Results and Interpretation

The visuals showed several patterns within the Airbnb dataset. The price distribution histogram showed that most listings fall between approximately \$50 and \$200 per night, with a smaller number of listings extending into much higher price ranges. This indicates a right-skewed distribution where most of the listings are moderately priced, while luxury listings are relatively rare.

The borough distribution visual showed that Manhattan and Brooklyn contain the largest number of Airbnb listings. Queens has a smaller number of listings, while the Bronx and Staten Island

contain a few properties. These patterns likely reflect differences in tourism demand, population density, and neighborhood attractiveness.

The scatter plot comparing price and number of reviews suggested that listings with very high prices tend to have fewer reviews, while moderately priced listings receive a larger number of reviews. This may indicate that moderately priced listings are booked more frequently.

Responsible Practice (Bias and Data Quality)

Data cleaning decisions can introduce bias if certain types of listings are removed from the dataset. For example, removing high price outliers may exclude legitimate luxury listings that represent a specific market segment. Similarly, missing review data may represent newly listed properties rather than missing information.

To reduce the risk of misleading results, cleaning steps were limited to clearly problematic values, like missing identifiers and extreme price values that could distort the analysis. Future analysis could incorporate additional variables such as property size, amenities, and geographic location to better understand pricing differences.

Reproducibility

This project was designed with reproducibility in mind. The analysis was conducted using a Jupyter Notebook that integrates code, narrative explanation, and visuals in a single document. Computational notebooks allow researchers to document each step of the data analysis process, including data cleaning, exploration, and visualization. This approach helps create transparent workflows that can be easily reproduced and shared with others (Danchev, 2022).

In addition, a requirements.txt file was generated to capture the exact Python package environment used in the project. The project was also managed using Git and GitHub with

multiple commits and a development branch, demonstrating version control practices that support reproducible research.

References

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, 5(56), 156. <https://doi.org/10.21105/jose.00156>